

Session 6

Task 4

- An **entry-controlled loop** checks its condition **before** executing the loop body, while an **exit-controlled loop** checks its condition **after** executing the body.
- Here is how each behaves when the condition is false right at the start.

1. Entry-Controlled Loops (for, while)

- The condition is checked upfront. If it evaluates to false on the very first check, the code inside the loop is skipped completely and executes **0 times**.

Ex.

```
#include <stdio.h>

main() {

    int count = 5;

    while (count < 3) {

        printf("This line will never print.\n");

        count++;

    }

}
```

2. Exit-Controlled Loop (do-while)

- The condition is checked at the bottom. Because the program enters the loop block before making its first check, the body executes **at least once**, even if the condition is false from the beginning.

Ex.

```
#include <stdio.h>

int main() {

    int count = 5;

    do {

        printf("This line prints exactly once!\n");

        count++;

    }
```

```
while (count < 3); // Condition (6 < 3) is false  
}
```